

Model Question Paper I**ELECTRICITY GENERATION, TRANSMISSION & DISTRIBUTION***Time: 3 Hour**Max.Marks: 75***PART A**

I. Answer **all** questions in one word or one sentence. Each question carries 1 mark.

1	A condenser in a thermal power plant condenses steam coming out of _____.	M1.01	R
2	State main function of moderator in a nuclear power plant.	M1.01	R
3	What is a synchronous condenser?	M2.03	R
4	Define load factor.	M2.01	R
5	Draw the power triangle.	M2.03	R
6	When the length of an overhead transmission line is about 50 km and the line voltage is comparatively low (< 20 kV), it is _____ transmission line	M3.01	R
7	List the standard transmission voltages used in India.	M3.01	R
8	_____ is the main consideration in the design of a feeder	M4.01	R
9	_____ is the main consideration in the design of a distributor	M4.01	R

PART B

II. Answer any **eight** questions from the following, each question carries 3 marks.

1	Outline factors to be considered for the selection of site for hydro electric power station.	M1.02	U
2	Discuss basic concept of micro grid.	M1.04	U
3	Calculate annual bill of a consumer whose maximum demand is 100kW, PF=0.8 lagging and load factor =60%.The tariff used is Rs 75 per KVA of maximum demand plus 15 paise per kWH consumed.	M2.04	A
4	List the disadvantages of low power factor.	M2.03	R
5	A generating station has a connected load of 43 MW and a maximum demand of 20MW, the units generated being 61.5×10^6 per annum. Calculate load factor.	M2.01	A
6	A transmission line has a span of 150 m between level supports. The tension in the conductor is 2000 kg.the weight of the conductor is 1.98kg/m wind pressure is 1.5 kg/m. calculate sag.	M3.02	A
7	A 132 kV transmission line has the following data: Wt. of conductor 0.680 kg/m, length of span 260 m, working tension 1550 kg. Calculate the height above ground at which the conductor should be supported. Ground clearance required is 10 metres.	M3.02	A
8	Summarize benefits of Flexible AC Transmission system.	M3.04	U
9	Discuss causes of insulator failure in the power system.	M4.02	U
10	Explain ring main system.	M4.01	U

PART C

Answer all questions. Each question carries 7 marks.

III	A hydro-electric generating station is supplied from a reservoir of capacity 5×10^6 cubic metre at a head of 200 metre. Find the total energy available in kWh if the overall efficiency is 75%.	M1.03	A
	OR		
IV	A hydroelectric station has an average available head of 100 metres and reservoir capacity of 50×10^6 cubic metres. Calculate the total energy in kWh that can be generated, assuming hydraulic efficiency of 85% and electrical efficiency of 90%.	M1.03	A
V	Describe a nuclear power station with the help of neat schematic diagram.	M1.01	U
	OR		
VI	Describe a coal based power station with the help of a neat schematic diagram.	M1.01	U
VII	Compare two part tariff and three part tariff.	M2.04	U
	OR		
VIII	Explain time of day (TOD) tariff.	M2.04	U
IX	Explain types of conductors used in overhead transmission	M3.01	U
	OR		
X	Discuss transposition of conductors and its necessity	M3.03	U
XI	Demonstrate layout of interconnected AC distribution system.	M4.01	U
	OR		
XII	Discuss classifications of cables based on voltage for which they are manufactured.	M4.03	U
XIII	Describe any one method of underground cable laying.	M4.03	U
	OR		
XIV	Identify classifications of distribution sub station.	M4.04	U

Model Question Paper II**ELECTRICITY GENERATION, TRANSMISSION & DISTRIBUTION***Time: 3 Hour**Max.Marks: 75***PART A**

III. Answer **all** questions in one word or one sentence. Each question carries 1 mark.

1	State the function of economizer in a steam power station.	M1.01	U
2	State the function of moderator in nuclear power plant	M1.01	R
3	what is demand factor.	M2.01	R
4	Define load factor.	M2.01	R
5	State voltage regulation.	M3.01	R
6	Define Ferranti effect.	M3.03	R
7	Show the expression for determining the string efficiency.	M4.02	U
8	Compare design factor of feeder and distributor.	M4.01	U
9	What is grading in underground cable	M4.03	R

PART B

IV. Answer any **eight** questions from the following, each question carries 3 marks.

1	List different conventional and non-conventional methods of power generation	M1.01	R
2	Explain main components of an atomic power plant .	M1.01	U
3	The maximum demand of a consumer is 10 A at 240 V and his total energy consumption is 8760 kWh. If the energy is charged at the rate of 8 rupees per unit for 300 hours use of the maximum demand per annum plus 2 rupees per unit for additional units, calculate: (i). annual bill (ii) equivalent flat rate. Assume the load factor and power factor to be unity.	M2.04	A
4	Explain the terms. (i). Load factor. (ii). Diversity factor.	M2.01	U
5	Explain the necessity of transposition in OH supply lines.	M3.03	U
6	Summarize different types of FACTS controllers	M3.04	U
7	Explain skin effect and factors affecting skin effect.	M3.03	U
8	Summarize flashover and puncture of insulators	M4.02	U
9	Show the construction of underground cable with suitable diagrams.	M4.03	R
10	Explain feeders, Distributors and service mains in an electric supply system	M4.01	U

PART C

Answer ALL questions. Each question carries 7 marks.

III	Define hydroelectric power station with the help of a neat schematic diagram.	M1.01	R
	OR		
IV	List the essential equipment in a steam power station and state the functions of each.	M1.01	R
V	A hydroelectric power plant operates under an effective head of 100m .Find the flow rate of water if power developed is 20MW and overall efficiency of the power plant is 70%.	M1.03	A
	OR		
VI	Solve for annual average power generated for a hydroelectric plant from the given data? catchment area - 10sqkm annual rainfall - 150cm expected Head - 200m expected yield factor - 60% power plant efficiency - 70% expected load factor - 0.6	M1.03	A
VII	Interpret base load and peak load with the help of load curve.	M2.01	U
	OR		
VIII	Explain importance ,effects and different methods of power factor improvement.	M2.03	U
IX	Illustrate various costs involved when determining total cost of electrical energy Generated.	M2.02	U
	OR		
X	Explain different types of tariff systems used to determine the cost of electric energy.	M2.04	U
XI	A transmission line has a span of 210 meters between level supports. The conductor has a cross-sectional area of 1:3 cm ² , weighs 1270 kg/km and has a breaking stress of 4228 kg/ cm ² . Calculate the sag for a safety factor of 6, allowing a wind pressure of 132 kg per square metre of projected area. Determine the slant sag.	M3.02	A
	OR		
XII	.A 132 kV transmission line has the following data: Wt. of conductor 680 kg/km, length of span 260 m, ultimate strength = 3100 kg , and safety factor = 2. Calculate the height above ground at	M3.02	A

	which the conductor should be supported. Ground clearance required is 10 metres		
XIII	Explain the method employed for laying underground cable with the help of neat diagrams	M4.03	U
	OR		
XIV	Explain the methods used for grading in underground cables	M4.03	U